

Week 8

Risk and Return

This week will be the answer for all of those questions:

Measuring returns and risks - SLO 3

Systematic and unsystematic risks - SLO 3

Portfolios and diversification - SLO 3

The Capital Asset Pricing Model (CAPM) - SLO 3

Learning Objectives:

At the end of this module you will:

SLO3: demonstrate an understanding of financial risk, its measurement, and its relationship to return

What are returns?

- The returns from an investment are the **future cash flows**.
- The return should **compensate** the investor for
 - **Time** (opportunity cost)
 - **Inflation**
 - **Risk**
 -
- **Every** investor wants compensation for at least time & inflation.
- Expected return increases due to increased risk
- In finance, **uncertainty is risk**

Calculating Estimated Returns

- We can measure an estimated return using start and end values

$$R_1 = \frac{P_1 - P_0}{P_0}$$

- But, some investments have periodic cash flows (e.g., dividends or coupon payments).
- The estimated return would be:

$$R_1 = \frac{(P_1 - P_0) + D_1}{P_0}$$

Return Statistics

- We actually experienced on a stock during a period the realized rate of return for that period
- However, the risk-return that investors face is **not** based on realized rates of return, it is based on what the investor expects to earn on an investment.
- To describe the range of possible returns, investors use the **expected rate of return**.

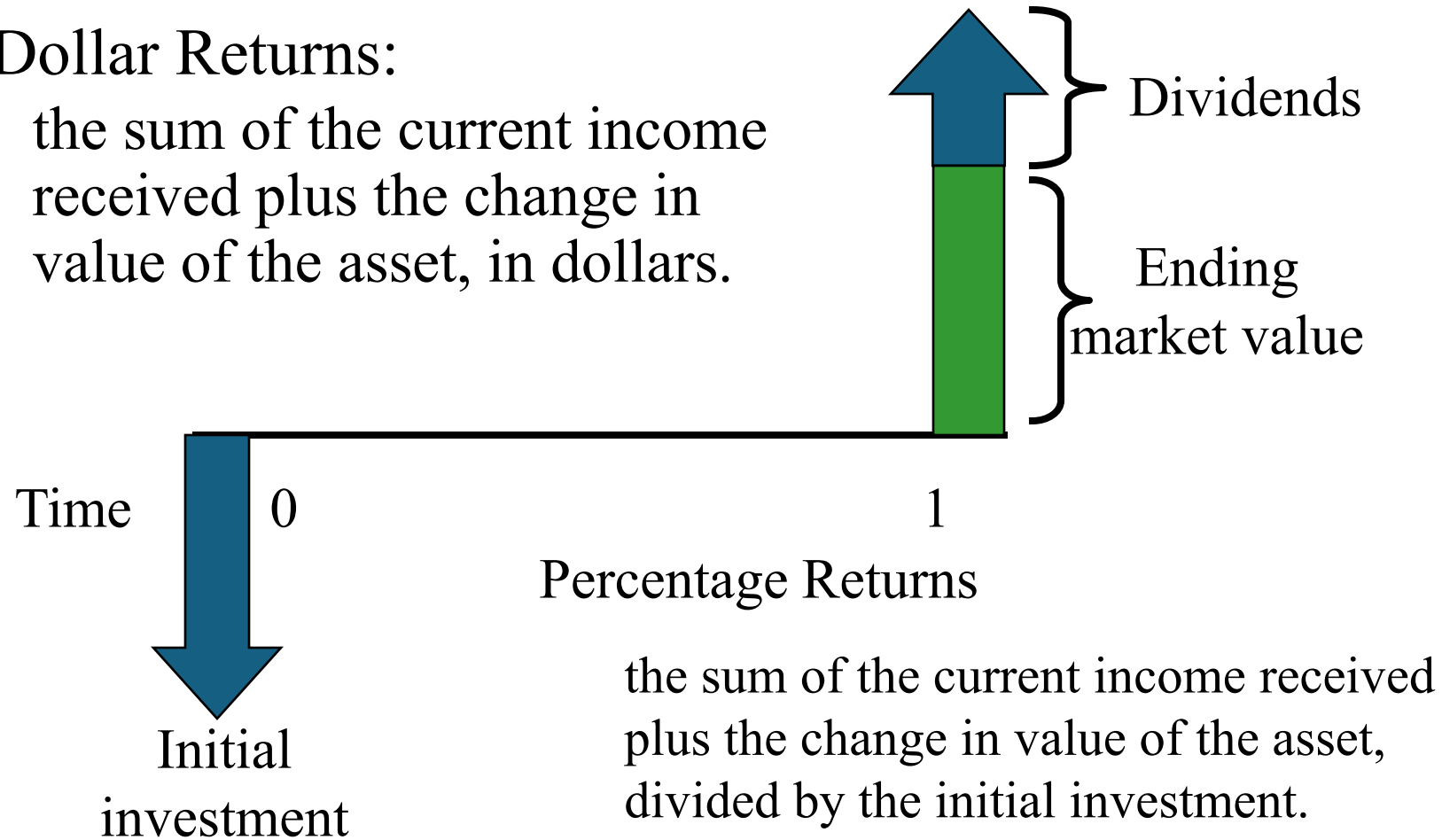
Calculating Expected (Future) Returns

- Future returns can be different under different conditions such as a boom or recession.
- To calculate an expected return we must include all possible outcomes weighting them on their probability.
- Expected Return = $\Sigma \text{Probability of Return} \times \text{Return}$
 - $E(r) = \Sigma P_i \times R_i$
- where
- $P_i = \text{Probability of outcome } i$
- $R_i = \text{Return for outcome } i$
-

Graphic Representation of Returns

► Dollar Returns:

the sum of the current income received plus the change in value of the asset, in dollars.



Formulaic Definition of Returns

Dollar Return = Dividend + Change in Market Value

$$\text{Percentage Return} = \frac{\text{dollar return}}{\text{beginning market value}}$$

$$= \frac{\text{dividend} + \text{change in market value}}{\text{beginning market value}}$$

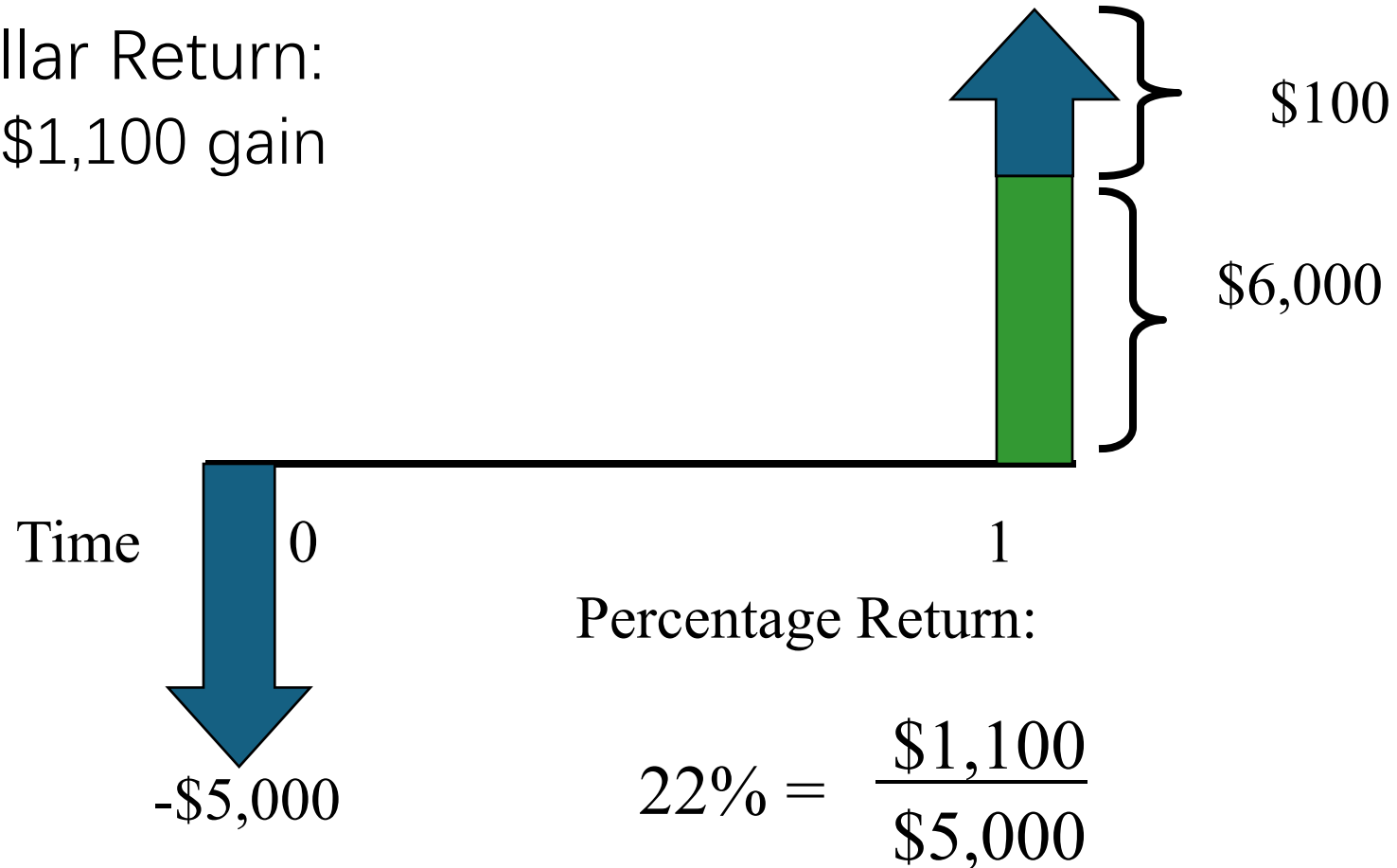
$$= \text{dividend yield} + \text{capital gains yield}$$

Example 1

- Suppose you bought 100 shares of Wal-Mart (WMT) one year ago today at \$50. Over the last year, you received \$100 in dividends (\$1 per share $\times$ 100 shares). At the end of the year, the stock sells for \$60. How did you do?
- Quite well. You invested $\$50 \times 100 = \$5,000$. At the end of the year, you have stock worth \$6,000 and cash dividends of \$100. Your dollar gain was $\$1,100 = \$100 + (\$6,000 - \$5,000)$.
- Your percentage gain for the year is:
 - $1,100/5,000 = 22\%$

Example 1

Dollar Return:
\$1,100 gain



Example 2

- Consider the following information, What is the expected return of the stock:

State of economy	boom	good	poor	bust
Probability of State of Economy	0.30	0.40	0.25	0.05
Rate of Return If State Occurs	0.3	0.12	0.1	-0.06

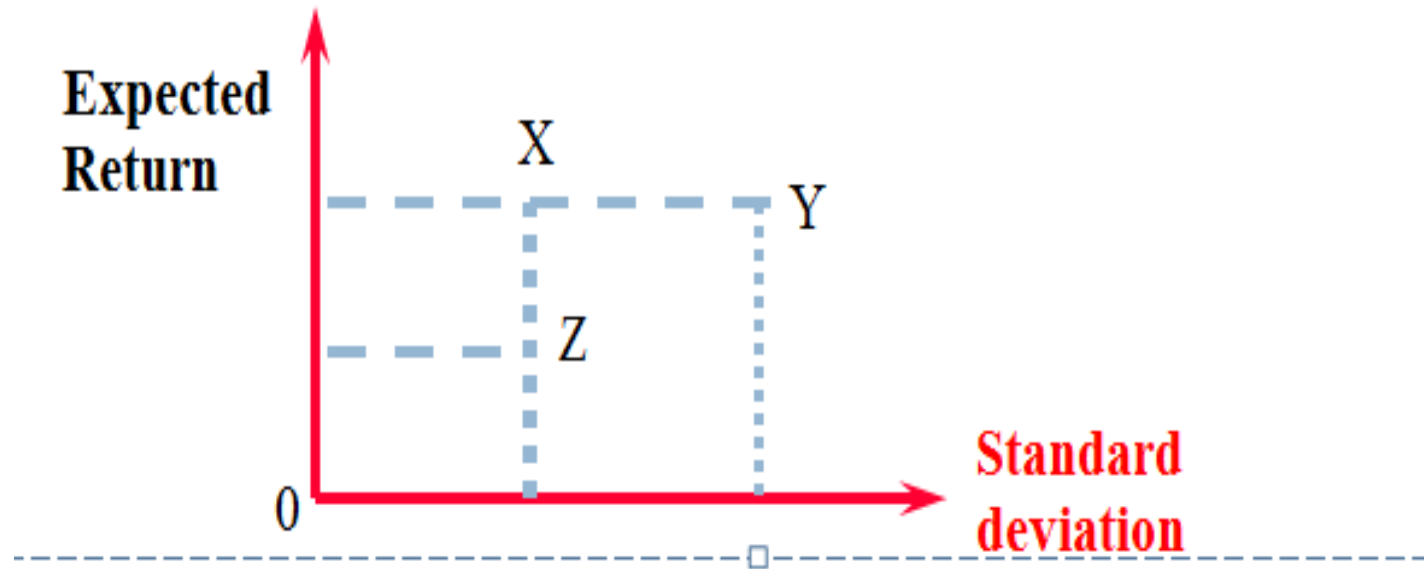
$$E(R_p) = .30(.3) + .40(.12) + .25(.01) + 0.05(-0.06) = .13750 \text{ or } 13.75\%$$

Variability of Returns

- The risk of an asset is the variability of its returns.
- More varied the returns relative to the average return the riskier the asset.
- Standard deviation measures how far the returns deviate from the average
- A risk-free asset has a standard deviation equal to zero.

Risk Aversion

- Assume all investors are risk-averse
- Demand higher return for higher risk
- If two investments have the same expected return
- Choose the one with the lower standard deviation



- In this example,
 - X is better than Y as we achieve the same return but X has lower Standard deviation(risk) than Y
 - X is better than Z as we achieve more return with X at the same level of Standard deviation(risk) as Z

Variance and Standard Deviation: Example

Coin Toss Game: calculating variance and standard deviation
(assume a mean of 10)

(1)	(2)	(3)
Percent Rate of Return	Deviation from Mean	Squared Deviation
+ 40	+ 30	900
+ 10	0	0
+ 10	0	0
- 20	- 30	900

Variance = average of squared deviations = $1800 / 4 = 450$

Standard deviation = square of root variance = $\sqrt{450} = 21.2\%$

Example 3

Consider the following two risky assets. There is a 1/3 chance of each state of the economy, and the only assets are a stock fund and a bond fund.

<i>Scenario</i>	<i>Probability</i>	<i>Rate of Return</i>	
		<i>Stock Fund</i>	<i>Bond Fund</i>
Recession	33.3%	-7%	17%
Normal	33.3%	12%	7%
Boom	33.3%	28%	-3%

Example 3

Scenario	Stock Fund		Bond Fund	
	Rate of Return	Squared Deviation	Rate of Return	Squared Deviation
<i>Recession</i>	-7%	0.0324	17%	0.0100
<i>Normal</i>	12%	0.0001	7%	0.0000
<i>Boom</i>	28%	0.0289	-3%	0.0100
Expected return	11.00%		7.00%	
Variance	0.0205		0.0067	
Standard Deviation	14.3%		8.2%	

Example 3

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Variance	0.0205		0.0067	
Standard Deviation	14.3%		8.2%	

$$E(r_s) = \frac{1}{3} \times (-7\%) + \frac{1}{3} \times (12\%) + \frac{1}{3} \times (28\%)$$

$$E(r_s) = 11\%$$

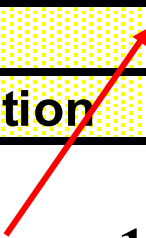
Example 3

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$$(-7\% - 11\%)^2 = .0324$$

Example 3

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$$.0205 = \frac{1}{3} (.0324 + .0001 + .0289)$$

Example 3

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$$14.3\% = \sqrt{0.0205}$$

Example 3: Covariance

<i>Scenario</i>	<i>Stock Deviation</i>	<i>Bond Deviation</i>	<i>Product</i>	<i>Weighted</i>
<i>Recession</i>	-18%	10%	-0.0180	-0.0060
<i>Normal</i>	1%	0%	0.0000	0.0000
<i>Boom</i>	17%	-10%	-0.0170	-0.0057
Sum				-0.0117
Covariance				-0.0117

Deviation compares return in each state to the expected return.

Weighted takes the product of the deviations multiplied by the probability of that state.

Example 3: Correlation

$$\rho = \frac{\text{Cov}(a, b)}{\sigma_a \sigma_b}$$

$$\rho = \frac{-.0117}{(.143)(.082)} = -0.998$$

Example 3 :

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Note that stocks have a higher expected return than bonds and higher risk. Let us turn now to the risk-return tradeoff of a portfolio that is 50% invested in bonds and 50% invested in stocks.

Example 3: Portfolio

<i>Scenario</i>	<i>Rate of Return</i>		<i>Portfolio</i>	<i>Squared deviation</i>
	<i>Stock fund</i>	<i>Bond fund</i>		
<i>Recession</i>	-7%	17%	5.0%	0.0016
<i>Normal</i>	12%	7%	9.5%	0.0000
<i>Boom</i>	28%	-3%	12.5%	0.0012
<i>Expected return</i>	11.00%	7.00%	9.0%	
<i>Variance</i>	0.0205	0.0067	0.0010	
Standard Deviation	14.31%	8.16%	3.08%	

The rate of return on the portfolio is a weighted average of the returns on the stocks and bonds in the portfolio:

$$r_P = w_B r_B + w_S r_S$$

$$5\% = 50\% \times (-7\%) + 50\% \times (17\%)$$

Example 3 : Portfolio

<i>Scenario</i>	<i>Rate of Return</i>		<i>Portfolio</i>	<i>Squared deviation</i>
	<i>Stock fund</i>	<i>Bond fund</i>		
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<i>Variance</i>	0.0205	0.0067	0.0010	
<i>Standard Deviation</i>	14.31%	8.16%	3.08%	

The *expected* rate of return on the portfolio is a weighted average of the *expected* returns on the securities in the portfolio.

$$E(r_P) = w_B E(r_B) + w_S E(r_S)$$

$$9\% = 50\% \times (11\%) + 50\% \times (7\%)$$

Example 3 : Portfolio

<i>Scenario</i>	<i>Rate of Return</i>		<i>Portfolio</i>	<i>Squared deviation</i>
	<i>Stock fund</i>	<i>Bond fund</i>		
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<i>Variance</i>	0.0205	0.0067	0.0010	
<i>Standard Deviation</i>	14.31%	8.16%	3.08%	

The variance of the rate of return on the two risky assets portfolio is

$$\sigma_P^2 = (w_B \sigma_B)^2 + (w_S \sigma_S)^2 + 2(w_B \sigma_B)(w_S \sigma_S)\rho_{BS}$$

where ρ_{BS} is the correlation coefficient between the returns on the stock and bond funds.

Example 3: Portfolio

<i>Scenario</i>	<i>Rate of Return</i>		<i>Portfolio</i>	<i>Squared deviation</i>
	<i>Stock fund</i>	<i>Bond fund</i>		
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Standard Deviation	14.31%	8.16%	3.08%	

Observe the decrease in risk that diversification offers.

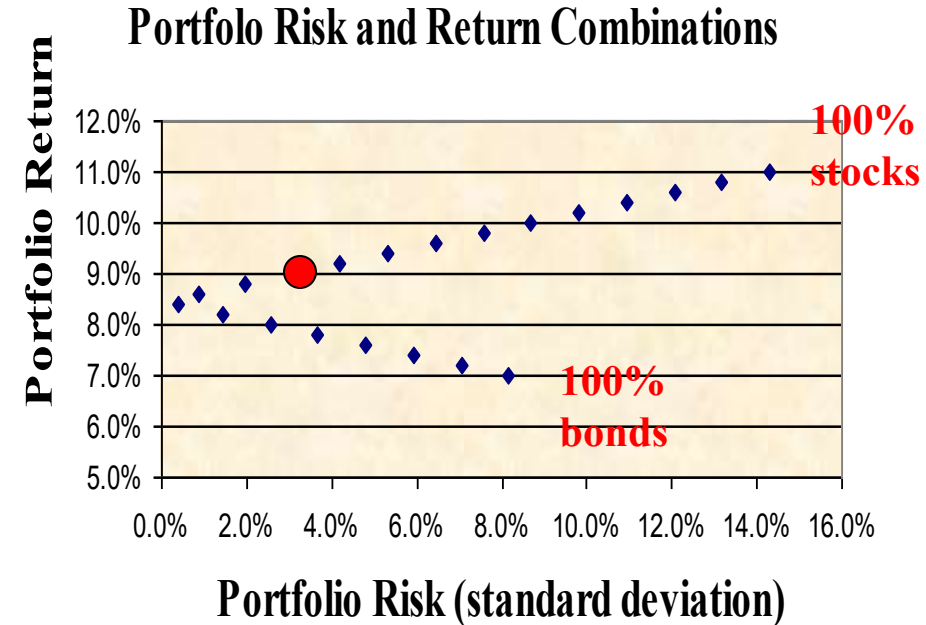
An equally weighted portfolio (50% in stocks and 50% in bonds) has less risk than either stocks or bonds held in isolation.

The efficient set

- The efficient set, or efficient frontier, is a graphical representation of a set of possible portfolios that:
 - Minimize risk at specific return levels; and,
 - Maximize returns at specific risk levels.

The Efficient Set: Example

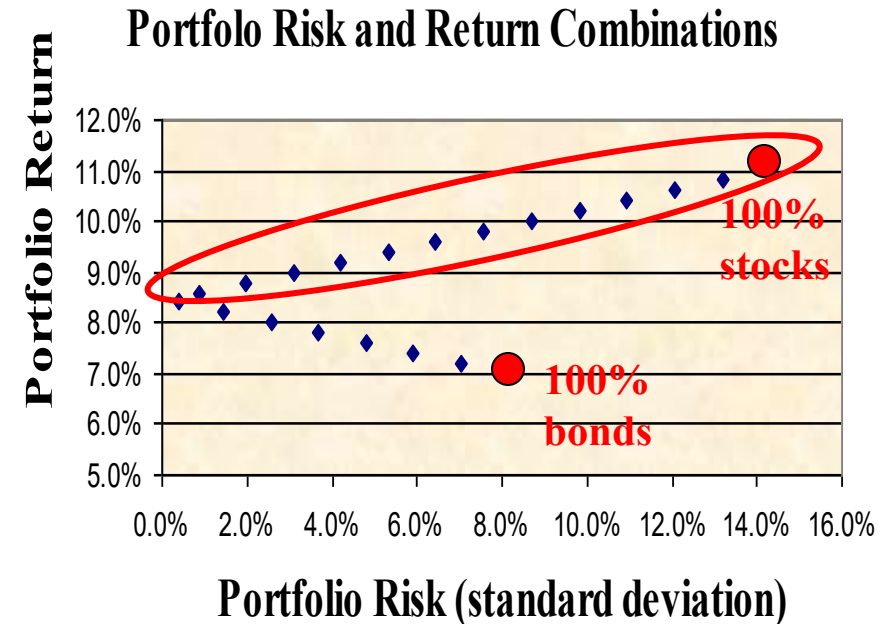
<i>% in stocks</i>	<i>Risk</i>	<i>Return</i>
0%	8.2%	7.0%
5%	7.0%	7.2%
10%	5.9%	7.4%
15%	4.8%	7.6%
20%	3.7%	7.8%
25%	2.6%	8.0%
30%	1.4%	8.2%
35%	0.4%	8.4%
40%	0.9%	8.6%
45%	2.0%	8.8%
50.00%	3.08%	9.00%
55%	4.2%	9.2%
60%	5.3%	9.4%
65%	6.4%	9.6%
70%	7.6%	9.8%
75%	8.7%	10.0%
80%	9.8%	10.2%
85%	10.9%	10.4%
90%	12.1%	10.6%
95%	13.2%	10.8%
100%	14.3%	11.0%



We can consider other portfolio weights besides 50% in stocks and 50% in bonds ...

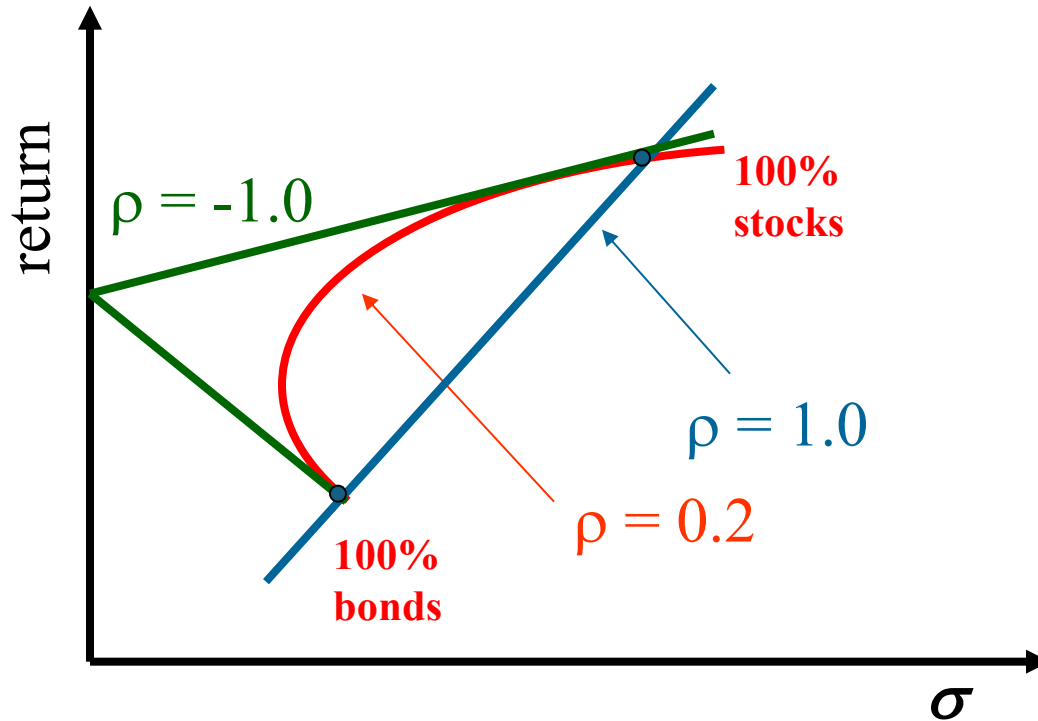
The Efficient Set: Example

<i>% in stocks</i>	<i>Risk</i>	<i>Return</i>
0%	8.2%	7.0%
5%	7.0%	7.2%
10%	5.9%	7.4%
15%	4.8%	7.6%
20%	3.7%	7.8%
25%	2.6%	8.0%
30%	1.4%	8.2%
35%	0.4%	8.4%
40%	0.9%	8.6%
45%	2.0%	8.8%
50%	3.1%	9.0%
55%	4.2%	9.2%
60%	5.3%	9.4%
65%	6.4%	9.6%
70%	7.6%	9.8%
75%	8.7%	10.0%
80%	9.8%	10.2%
85%	10.9%	10.4%
90%	12.1%	10.6%
95%	13.2%	10.8%
100%	14.3%	11.0%



Note that some portfolios are “better” than others. They have higher returns for the same level of risk (or less).

Portfolios with Various Correlations



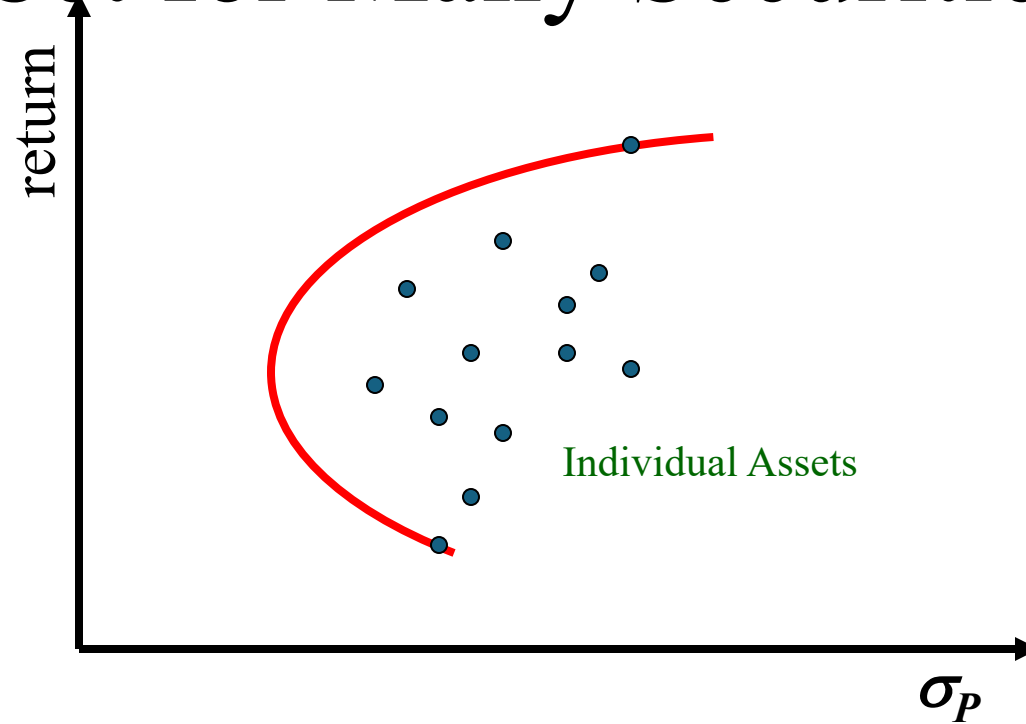
Relationship depends on correlation coefficient

$$-1.0 \leq \rho \leq +1.0$$

If $\rho = +1.0$, no risk reduction is possible

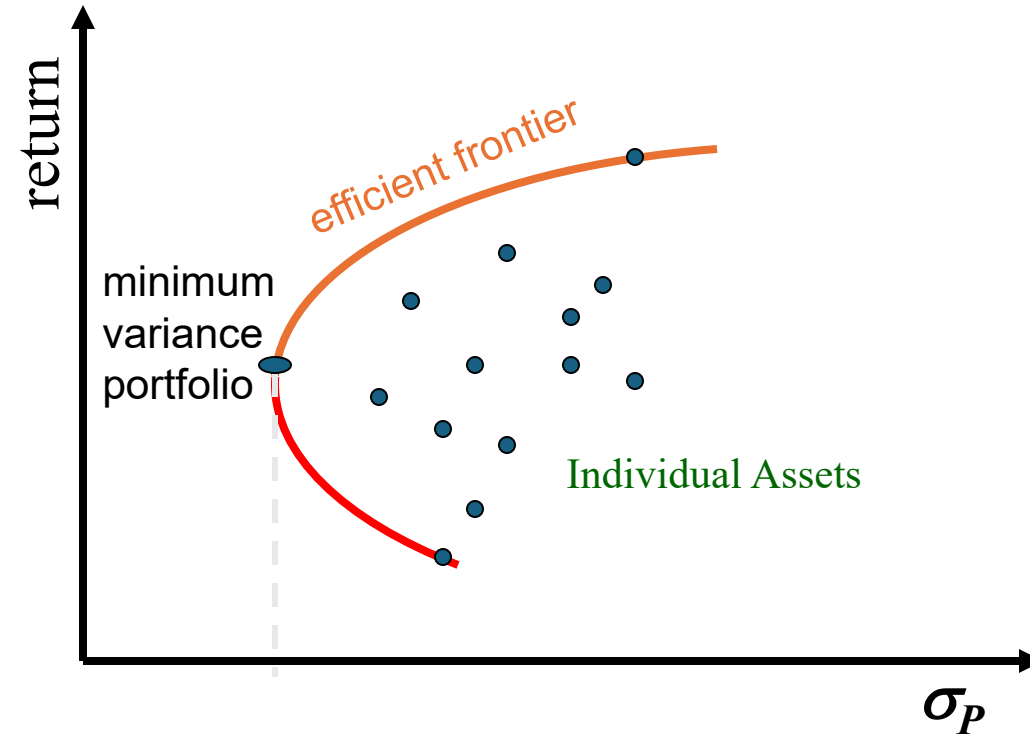
If $\rho = -1.0$, complete risk reduction is possible

The Efficient Set for Many Securities



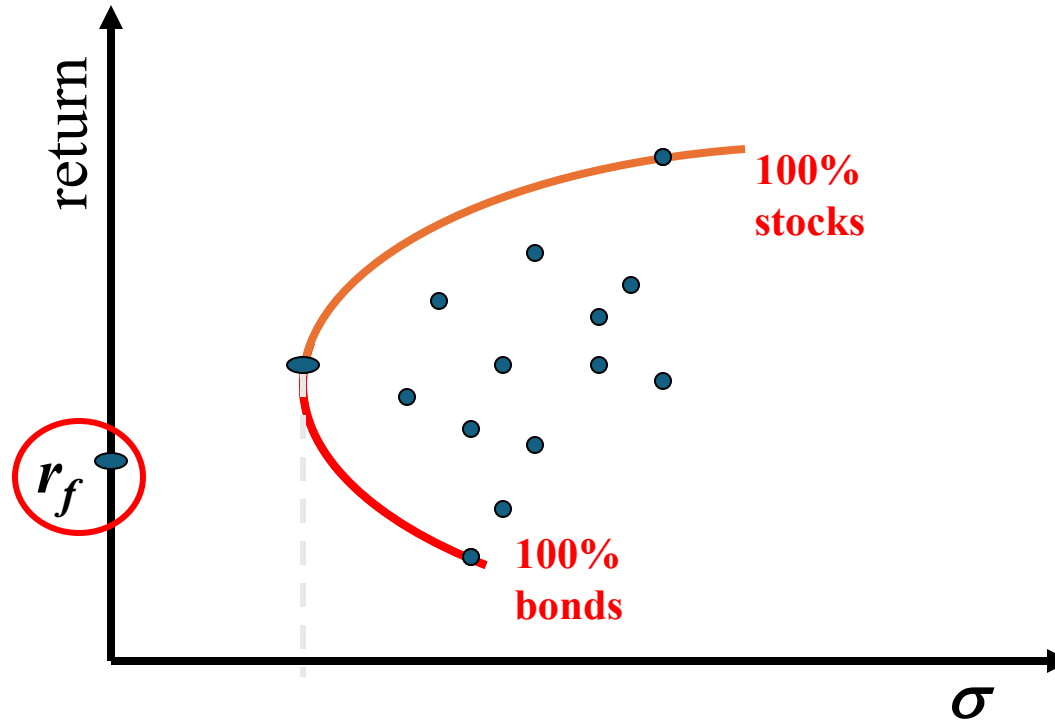
Consider a world with many risky assets; we can still identify the *opportunity set* of risk-return combinations of various portfolios.

The Efficient Frontier



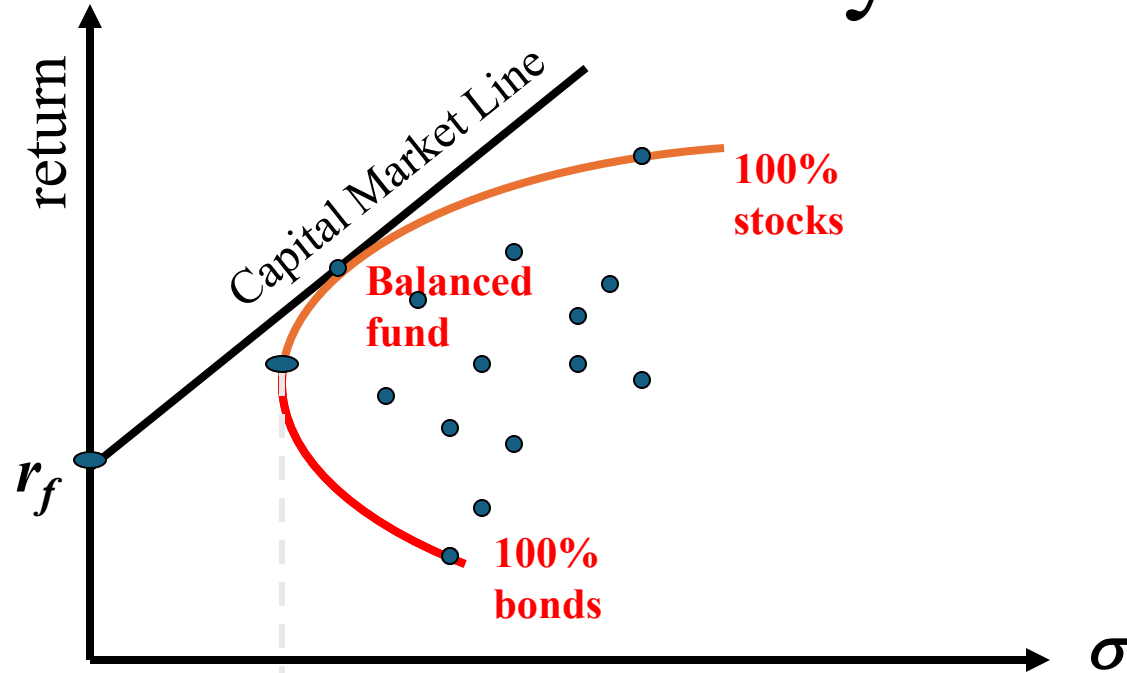
The section of the opportunity set above the minimum variance portfolio is the efficient frontier.

Riskless Borrowing and Lending



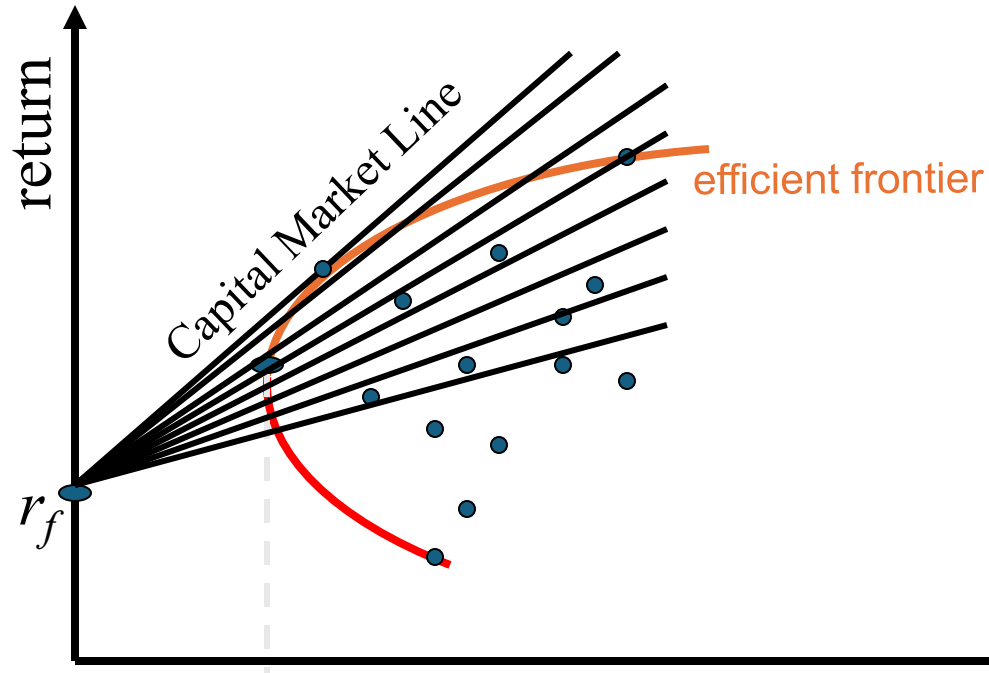
In addition to stocks and bonds, consider a world that also has risk-free securities like T-bills.

Impact of a Riskless Security



Now investors can allocate their money across the T-bills and a balanced mutual fund.

Optimal Portfolio with a Risk-Free Asset



With a risk-free asset available and the efficient frontier identified, we choose the capital allocation line with the steepest slope.

Risk: Systematic

- Risk factors that affect a large number of assets
- Also known as non-diversifiable risk or market risk
- Includes such things as changes in GDP, inflation, interest rates, etc.

Risk: Unsystematic

- Risk factors that affect a limited number of assets
- Also known as unique risk and asset-specific risk
- Includes such things as labor strikes, part shortages, etc.

Returns

- Total Return = expected return + unexpected return
- Unexpected return = systematic portion + unsystematic portion
- Therefore, total return can be expressed as follows:
 - Total Return = expected return + systematic portion + unsystematic portion

Diversification and Portfolio Risk

- Diversification can substantially reduce the variability of returns without an equivalent reduction in expected returns
- This reduction in risk arises because worse than expected returns from one asset are offset by better than expected returns from another
- However, there is a minimum level of risk that cannot be diversified away, and that is the systematic portion

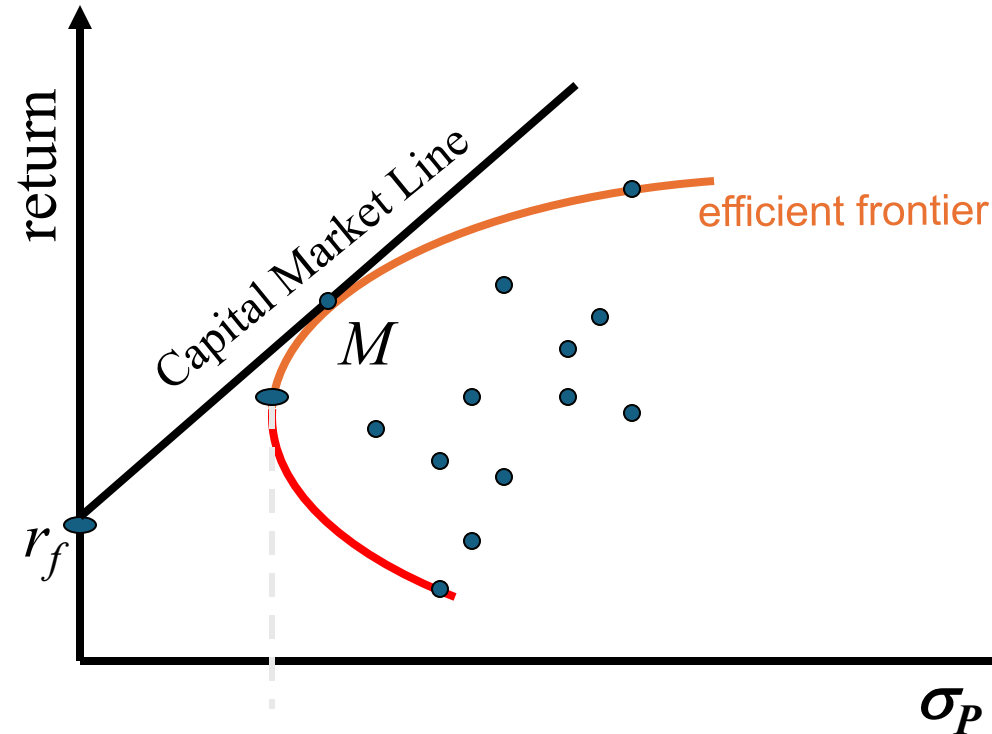
Diversifiable Risk

- The risk that can be eliminated by combining assets into a portfolio
- Often considered the same as unsystematic, unique, or asset-specific risk
- If we hold only one asset, or assets in the same industry, then we are exposing ourselves to risk that we could diversify away

Total Risk

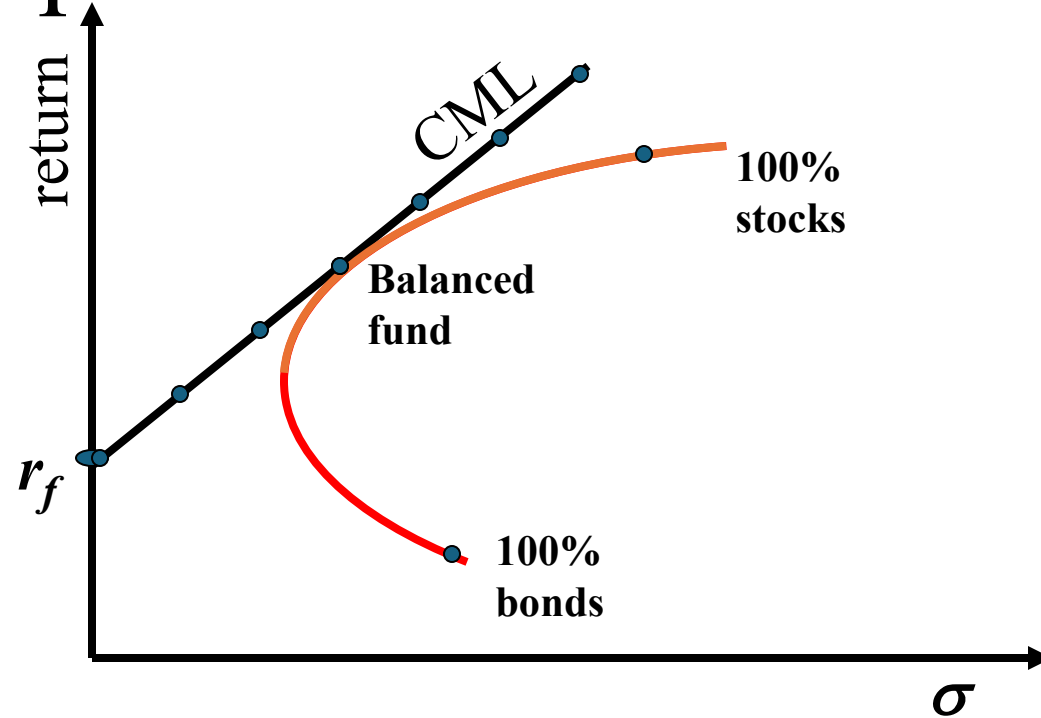
- Total risk = systematic risk + unsystematic risk
- The standard deviation of returns is a measure of total risk
- For well-diversified portfolios, unsystematic risk is very small
- Consequently, the total risk for a diversified portfolio is essentially equivalent to the systematic risk

Market Equilibrium



With the capital allocation line identified, all investors choose a point along the line—some combination of the risk-free asset and the market portfolio M . In a world with homogeneous expectations, M is the same for all investors.

Market Equilibrium



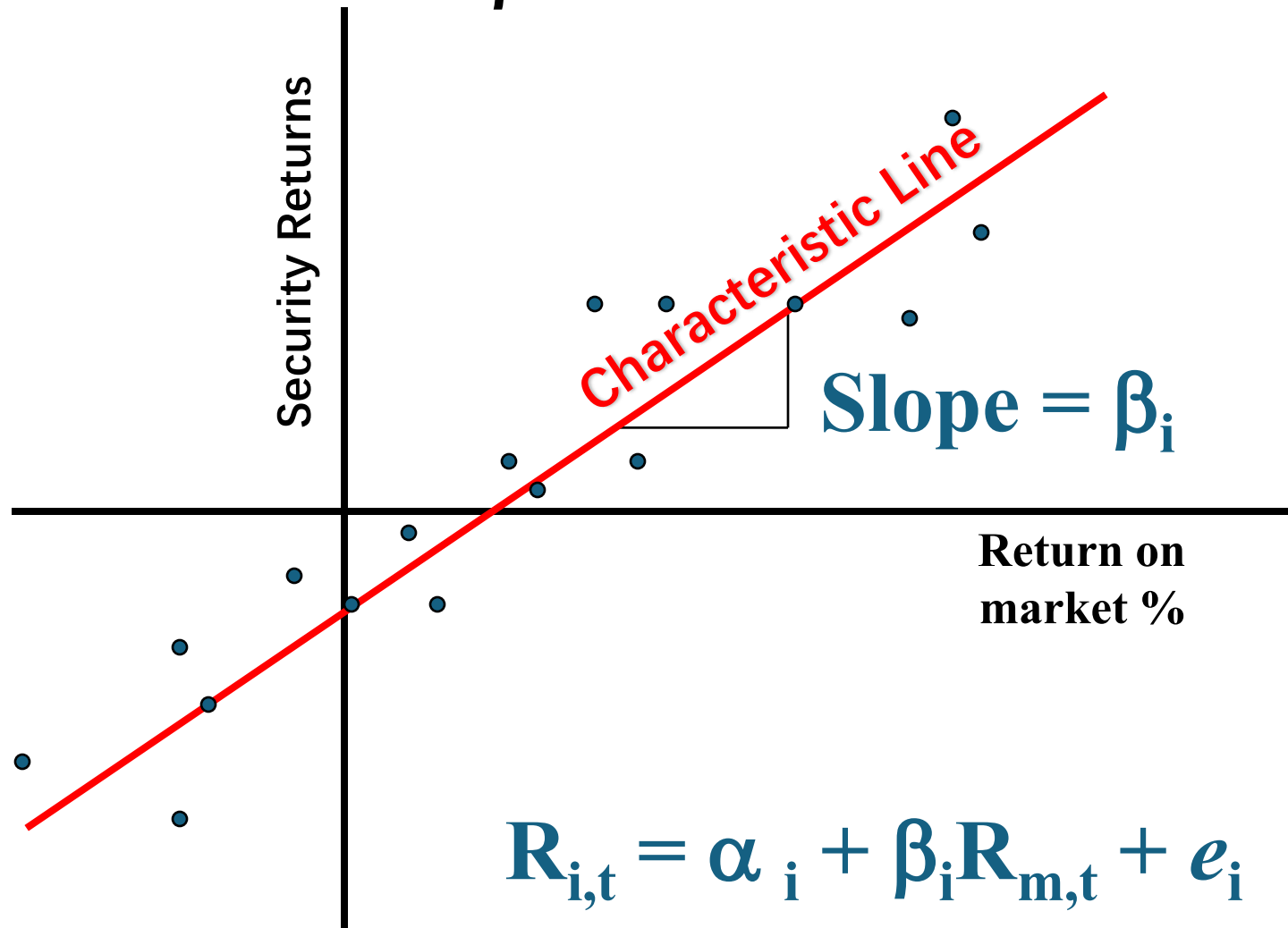
Just where the investor chooses along the Capital Market Line depends on his risk tolerance. The big point is that all investors have the same CML.

Risk When Holding the Market Portfolio

- Researchers have shown that the “best” measure of the risk of a security in a large portfolio is the *beta* (β) of the security
- Beta measures the responsiveness of a security to movements in the market portfolio (i.e., systematic risk)

$$\beta_i = \frac{Cov(R_i, R_M)}{\sigma^2(R_M)}$$

ESTIMATING β WITH REGRESSION



The Formula for Beta

$$\beta_i = \frac{Cov(R_i, R_M)}{\sigma^2(R_M)}$$

Clearly, your estimate of beta will depend upon your choice of a proxy for the market portfolio.

Relationship between Risk and expected Return (CAPM)

- Expected Return on the Market:

$$\bar{R}_M = R_F + \text{Market Risk Premium}$$

- Expected return on an individual security:

$$\bar{R}_i = R_F + \beta_i \times \underbrace{(\bar{R}_M - R_F)}$$

Market Risk Premium

This applies to individual securities held within well-diversified portfolios.

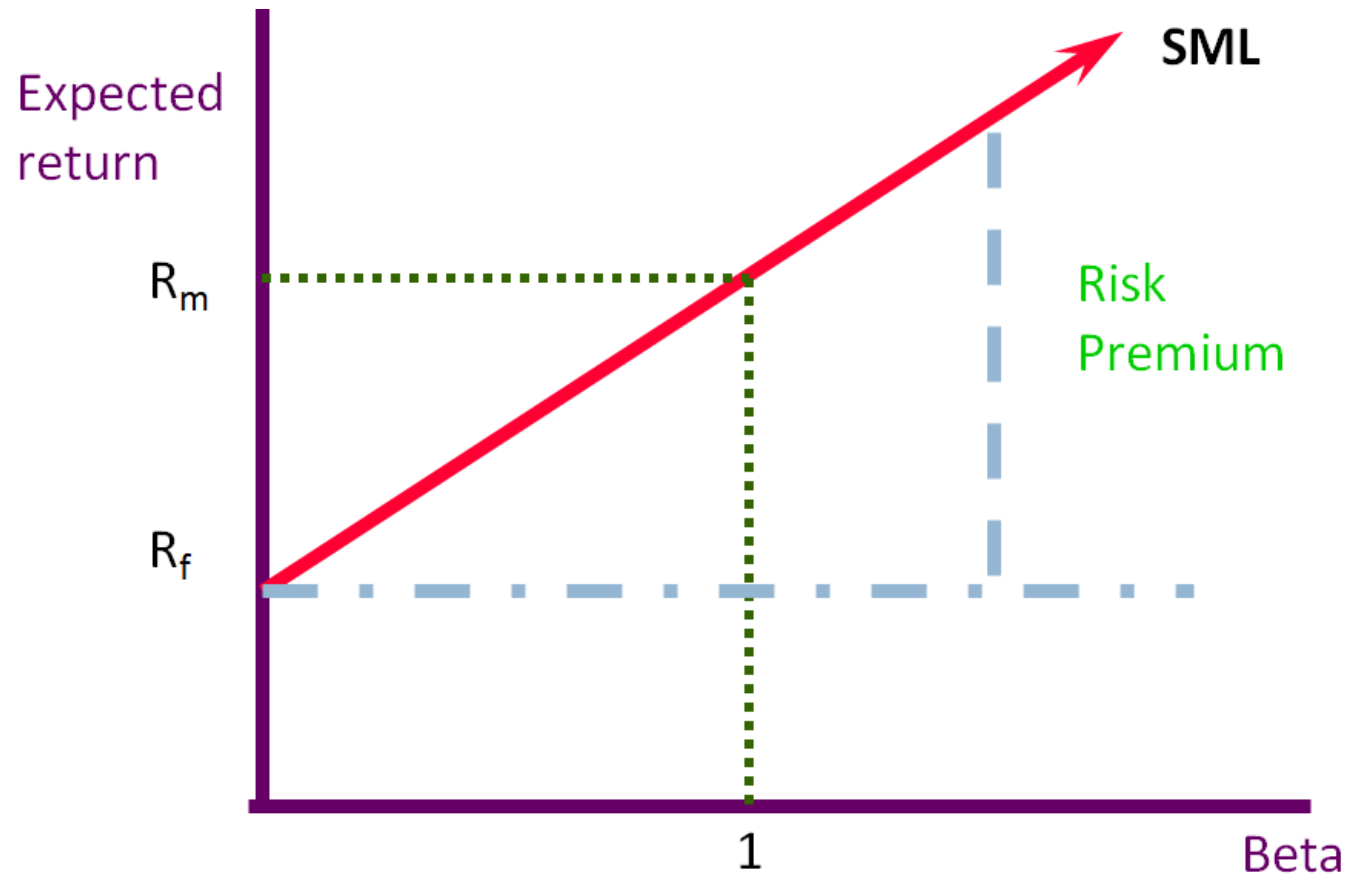
Expected Return on a Security

- This formula is called the Capital Asset Pricing Model (CAPM):

$$\bar{R}_i = R_F + \beta_i \times (\bar{R}_M - R_F)$$

Expected return on a security	=	Risk- free rate	+	Beta of the security	×	Market risk premium
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- Assume $\beta_i = 0$, then the expected return is R_F .
- Assume $\beta_i = 1$, then $\bar{R}_i = \bar{R}_M$



Practice 1

- Calculate the Beta of the portfolio below:

Security	Investment	Beta	Weight
ANZ	\$5,000	0.75	25%
ASX	\$7,000	1.10	35%
COH	\$8,000	1.36	40%

Advantages and Disadvantages of CAPM

Advantages

- ◆ it explicitly adjusts for risk
- ◆ can be used for companies that do not pay dividends and/or do not have constant growth

Disadvantages

- ◆ requires the market risk premium to be estimated
- ◆ need to be able to calculate beta
- ◆ rely on past observations to predict future

3. **Portfolio Expected Return** You own a portfolio that is 20 percent invested in Stock X, 35 percent invested in Stock Y, and 45 percent invested in Stock Z. The expected returns on these three stocks are 9.2 percent, 11.8 percent, and 14.3 percent, respectively. What is the expected return on the portfolio?

3. The expected return of a portfolio is the sum of the weight of each asset times the expected return of each asset. So, the expected return of the portfolio is:

$$E(R_p) = .20(.092) + .35(.118) + .45(.143)$$

$$E(R_p) = .1241, \text{ or } 12.41\%$$

11. Calculating Portfolio Betas You own a portfolio equally invested in a risk-free asset and two stocks. If one of the stocks has a beta of 1.17 and the total portfolio is equally as risky as the market, what must the beta be for the other stock in your portfolio?

11. The beta of a portfolio is the sum of the weight of each asset times the beta of each asset. If the portfolio is as risky as the market it must have the same beta as the market. Since the beta of the market is one, we know the beta of our portfolio is one. We also need to remember that the beta of the risk-free asset is zero. It has to be zero since the asset has no risk. Setting up the equation for the beta of our portfolio, we get:

$$\beta_p = 1.0 = 1/3(0) + 1/3(1.17) + 1/3(\beta_X)$$

Solving for the beta of Stock X, we get:

$$\beta_X = 1.83$$

18. Reward-to-Risk Ratios Stock Y has a beta of 1.20 and an expected return of 14.1 percent. Stock Z has a beta of .78 and an expected return of 9.5 percent. If the risk-free rate is 4.3 percent and the market risk premium is 7.6 percent, are these stocks correctly priced?

First, we can use the CAPM. Substituting in the values we are given for each stock, we find:

$$E(R_Y) = .043 + .076(1.20)$$

$$E(R_Y) = .1342, \text{ or } 13.42\%$$

It is given in the problem that the expected return of Stock Y is 14.1 percent, but according to the CAPM, based on its level of risk, the expected return should be 13.42 percent. This means the stock return is too low, given its level of risk. Stock Y plots above the SML and is undervalued. In other words, its price must increase to decrease the expected return to 13.42 percent. For Stock Z, we find:

$$E(R_Z) = .043 + .076(.78)$$

$$E(R_Z) = .1023, \text{ or } 10.23\%$$

The return given for Stock Z is 9.5 percent, but according to the CAPM the expected return of the stock should be 10.23 percent based on its level of risk. Stock Z plots below the SML and is overvalued. In other words, its price must decrease to increase the expected return to 10.23 percent.

23. Analyzing a Portfolio You want to create a portfolio equally as risky as the market, and you have \$1,000,000 to invest. Given this information, fill in the rest of the following table:

ASSET	INVESTMENT	BETA
Stock A	\$215,000	.83
Stock B	\$360,000	1.19
Stock C		1.30
Risk-free asset		

$$X_A = \$215,000 / \$1,000,000$$

$$X_A = .215$$

$$X_B = \$360,000 / \$1,000,000$$

$$X_B = .36$$

$$\beta_p = 1.0 = X_A(.83) + X_B(1.19) + X_C(1.30) + X_{Rf}(0)$$

$$1.0 = .215(.83) + .36(1.19) + X_C(1.30)$$

$$X_C = .302423$$

$$1 = X_A + X_B + X_C + X_{Rf} = 1 - .215 - .36 - .302423 - X_{Rf}$$

$$X_{Rf} = .122577$$

Invest in risk-free asset

$$= .122577(\$1,000,000)$$

$$\text{Invest in risk-free asset} = \$122,576.92$$

c. How does the correlation between the returns on A and B affect the standard deviation of the portfolio?

29. Correlation and Beta You have been provided the following data on the securities of three firms, the market portfolio, and the risk-free asset:

SECURITY	EXPECTED RETURN	STANDARD DEVIATION	CORRELATION*	BETA
Firm A	.11	.36	(i)	.85
Firm B	.14	(ii)	.38	1.35
Firm C	.13	.42	.57	(iii)
The market portfolio	.12	.19	(iv)	(v)
The risk-free asset	.03	(vi)	(vii)	(viii)

*With the market portfolio.

- Fill in the missing values in the table.
- Is the stock of Firm A correctly priced according to the capital asset pricing model (CAPM)?
What about the stock of Firm B? Firm C? If these securities are not correctly priced, what is your investment recommendation for someone with a well-diversified portfolio?

- a. (i) Using the equation to calculate beta, we find:
 - $\beta_I = (\rho_{I,M})(\sigma_I)/\sigma_M$
 - $.85 = (\rho_{I,M})(.36)/.19$
 - $\rho_{I,M} = .45$
- (ii) Using the equation to calculate beta, we find:
 - $\beta_I = (\rho_{I,M})(\sigma_I)/\sigma_M$
 - $1.35 = (.38)(\sigma_I)/.19$
 - $\sigma_I = .68$
- (iii) Using the equation to calculate beta, we find:
 - $\beta_I = (\rho_{I,M})(\sigma_I)/\sigma_M$
 - $\beta_I = (.57)(.42)/.19$
 - $\beta_I = 1.26$
- (iv) The market has a correlation of 1 with itself.
- (v) The beta of the market is 1.
- (vi) The risk-free asset has zero standard deviation.
- (vii) The risk-free asset has zero correlation with the market portfolio.
- (viii) The beta of the risk-free asset is 0.

- *Firm A:*

- $E(R_A) = R_f + \beta_A[E(R_M) - R_f]$
- $E(R_A) = .03 + .85(.12 - .03)$
- $E(R_A) = .1065$, or 10.65

- *Firm B:*

- $E(R_B) = R_f + \beta_B[E(R_M) - R_f]$
- $E(R_B) = .03 + 1.35(.12 - .03)$
- $E(R_B) = .1515$, or 15.15%

- *Firm C:*

- $E(R_C) = R_f + \beta_C[E(R_M) - R_f]$
- $E(R_C) = .03 + 1.26(.12 - .03)$
- $E(R_C) = .1434$, or 14.34%

36. SML Suppose you observe the following situation:

STATE OF ECONOMY	PROBABILITY OF STATE	RETURN IF STATE OCCURS	
		STOCK A	STOCK B
Bust	.20	−.08	−.07
Normal	.60	.12	.12
Boom	.20	.46	.32

- Calculate the expected return on each stock.
- Assuming the capital asset pricing model holds and Stock A's beta is greater than Stock B's beta by .43, what is the expected market risk premium?

$$E(R_A) = .20(-.08) + .60(.12) + .20(.46)$$

$$E(R_A) = .1480, \text{ or } 14.80\%$$

$$E(R_B) = .20(-.07) + .60(.18) + .20(.32)$$

$$E(R_B) = .1220, \text{ or } 12.20\%$$

- *b.* We can use the expected returns we calculated to find the slope of the Security Market Line. We know that the beta of Stock A is .43 greater than the beta of Stock B. Therefore, as beta increases by .43, the expected return on a security increases by .026 (= .1480 – .1220). The slope of the security market line (SML) equals:

-

- $$\text{Slope}_{\text{SML}} = \text{Rise/Run}$$

- $$\text{Slope}_{\text{SML}} = \text{Increase in expected return/Increase in beta}$$

- $$\text{Slope}_{\text{SML}} = (.1480 - .1260)/.43$$

- $$\text{Slope}_{\text{SML}} = .0605, \text{ or } 6.05\%$$

- We could also solve this problem using CAPM. The equations for the expected returns of the two stocks are:

- $$.148 = R_f + (\beta_B + .43)(\text{MRP})$$

- $$.122 = R_f + \beta_B(\text{MRP})$$

- We can rewrite the CAPM equation for Stock A as:

- $$.148 = R_f + \beta_B(\text{MRP}) + .43(\text{MRP})$$

- Subtracting the CAPM equation for Stock B from this equation yields:

- $$.026 = .43\text{MRP}$$

- $$\text{MRP} = .0605, \text{ or } 6.05\%$$